

# Majid Khoshrou

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## PROFILE

Data Scientist with 10+ years of experience bridging academia and industry, tackling high-impact challenges in energy, mobility, and infrastructure. My expertise spans machine learning, time series forecasting, NLP, and Generative AI - from optimizing national energy forecasts to building Mr M, a domain-specific AI assistant powered by LLMs and Retrieval-Augmented Generation. I specialize in turning complex, heterogeneous data into clear, actionable insights, designing systems that are not only accurate and scalable, but also explainable and production-ready.

A passionate advocate for open-source innovation, I actively contribute to projects like OpenSTEF and develop independent AI tools for my own platform, [majidkhoshrou.com](http://majidkhoshrou.com). My work spans diverse domains - including energy systems, robotics, marine science, and decision intelligence - all connected by a commitment to applied, ethical AI that drives measurable societal and business impact.

## SKILLS

**Programming & ML:** Python, SQL, MATLAB, PySpark, Bash, scikit-learn, Pandas, NumPy, TensorFlow, PyTorch, XGBoost, LightGBM, spaCy

**Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), OpenAI APIs, LangChain, Prompt Engineering, Embedding Pipelines, FAISS Indexing

**Data & Cloud Platforms:** AWS, GCP, Azure, Databricks, Docker, Git

**Analytics & Visualization:** Power BI, Tableau, Matplotlib, Seaborn, Plotly

**Methods:** Time Series Forecasting, NLP, Risk Modeling, Probabilistic Modeling, Anomaly Detection

**Collaboration & Management:** Agile/Scrum, JIRA, Confluence, Stakeholder Engagement

**Open Source:** Contributor to [OpenSTEF](#) (Energy Forecasting Library)

**Languages:** English (Fluent), Dutch (Intermediate), Persian (Native)

## WORK EXPERIENCE

**Senior Data Scientist**, Alliander – Arnhem

Jan 2023 – Present

- Increased day-ahead allocation forecast accuracy by 30%, delivering over one million euros annual cost savings.
- Identified major cost-reduction opportunities via in-depth analysis of energy settlement pricing.
- Enhanced OpenSTEF open-source library, improving usability and model robustness.
- Developed statistical risk models to assess and mitigate grid reliability issues.
- Standardized model validation processes across teams, ensuring consistent performance tracking.

**Postdoctoral Researcher**, Centrum Wiskunde & Informatica – Amsterdam

Jul 2020 – Dec 2022

- Built accurate EV charging demand forecasts to guide urban mobility infrastructure planning.
- Introduced carbon impact metrics for server clusters, shaping sustainability reporting practices.
- Co-taught graduate-level courses in AI and game theory, improving student engagement and understanding.

**Data Scientist**, Maistering B.V. – Amsterdam/Rotterdam

Nov 2019 – Jul 2020

- Designed customer segmentation models that improved marketing ROI.

- Delivered ML-powered product features for enterprise clients, enhancing market value.
- Accelerated deployment of analytics pipelines through cross-team collaboration.

**PhD Researcher**, CWI & TU Delft – Netherlands

Dec 2015 – Nov 2019

- Developed probabilistic forecasting techniques that improved smart grid prediction reliability.
- Created anomaly detection methods to strengthen grid health monitoring.
- Published peer-reviewed research advancing time series analysis in energy systems.

**Machine Learning Researcher**, C2SR Lab – Porto, Portugal

May 2013 – Oct 2015

- Developed real-time learning algorithms that improved marine robotics navigation efficiency.
- Improved adaptive sampling strategies for more effective ocean mission data collection.

**Project Member**, EDSAB Co. – Tehran, Iran

Feb 2011 – Feb 2012

- Developed ML models to detect anomalies and potential fraud in national smart meter datasets, improving theft detection accuracy and reducing grid losses.
- Developed long-term energy demand forecasts for national infrastructure planning.

## AI PROJECTS

**Mr M - Domain-Specific Generative AI Assistant**

2025 – Present

- Designed and deployed a personalized AI assistant trained on my own academic and professional materials.
- Built a robust RAG pipeline using OpenAI embeddings and FAISS for semantic search with context-restricted QA.
- Implemented automated text extraction, chunking, and semantic embedding across heterogeneous sources.
- Ensured explainability and trustworthiness through traceable answers linked to original sources.

## EDUCATION

**PhD in Artificial Intelligence**, *CWI and Delft University of Technology*

2015 – 2022

Thesis: "Singular Value Decomposition for Time Series Analysis in Smart Energy Systems"

**MSc in Information Engineering**, *University of Porto, Portugal*

2012 – 2015

Thesis: "Real-Time Unsupervised Motion Learning for Autonomous Underwater Vehicles"

**BSc in Electrical Power Engineering**, *Babol Noshirvani University of Technology, Iran* 2002 – 2007

## CERTIFICATIONS

- [Google Advanced Data Analytics Certificate](#) (2024)
- [Google IT Support Specialization](#) (2021)
- [Statistics Fundamentals with Python Track](#) (2019)
- [Data Scientist with Python Track](#) (2019)

## ADDITIONAL INFORMATION

**Work Eligibility:** EU citizen (Dutch nationality), also holds Iranian citizenship

**Interests:** Fitness, photography, travel, museums, technology, chess